

Scaled Bridge – Version 1

04OAGQD - Vibration Mechanics

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1. Accelerometer Positions

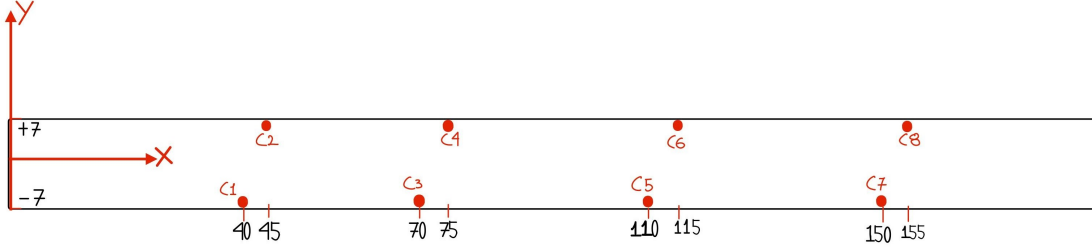


Figure 1: Positions of accelerometers on the scaled bridge.

Eight accelerometers (C1–C8) were placed along the bridge span at precise distances in cm: C1 at 40, C2 at 45, C3 at 70, C4 at 75, C5 at 110, C6 at 115, C7 at 150, and C8 at 155. Odd-numbered channels are on the bottom chord ($Y = -7$ cm), and even-numbered ones on the top chord ($Y = +7$ cm). This symmetric arrangement captures both vertical and flexural vibrations with spatial resolution across the span. The distances along x-axis of the accelerometers were chosen carefully not to be on the nodes.

2. Auto-PSD of All Channels

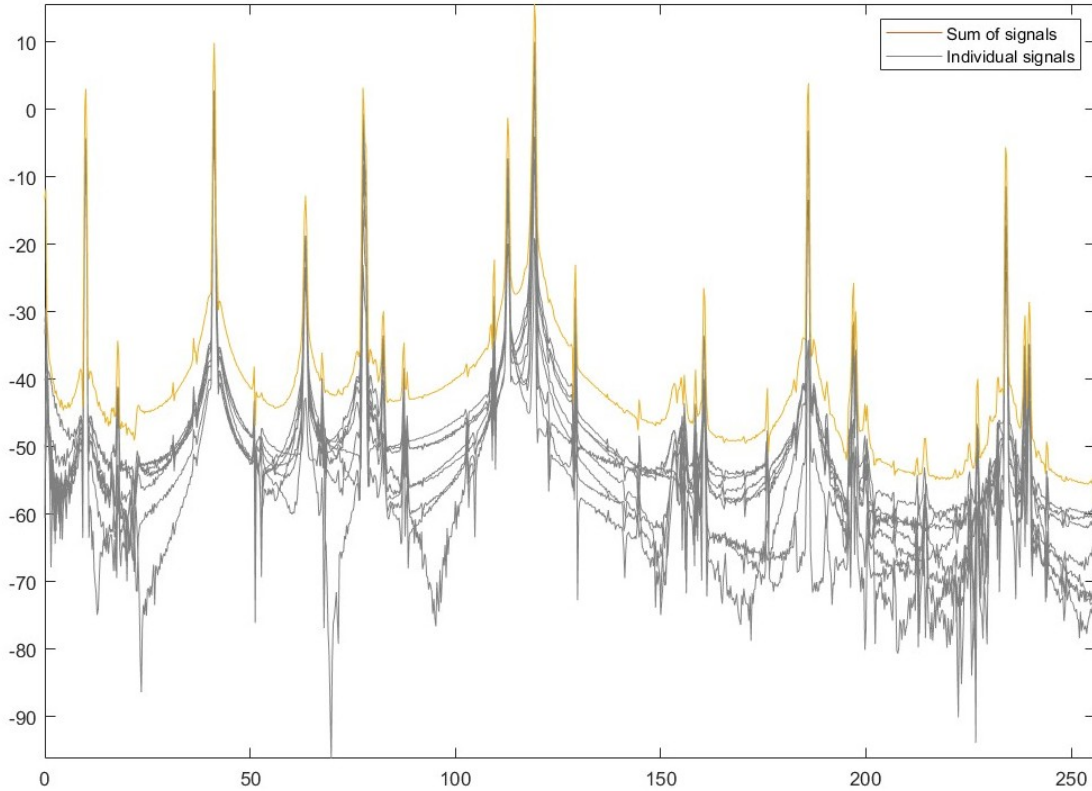


Figure 2: Summed auto-PSD across all acquisition channels.

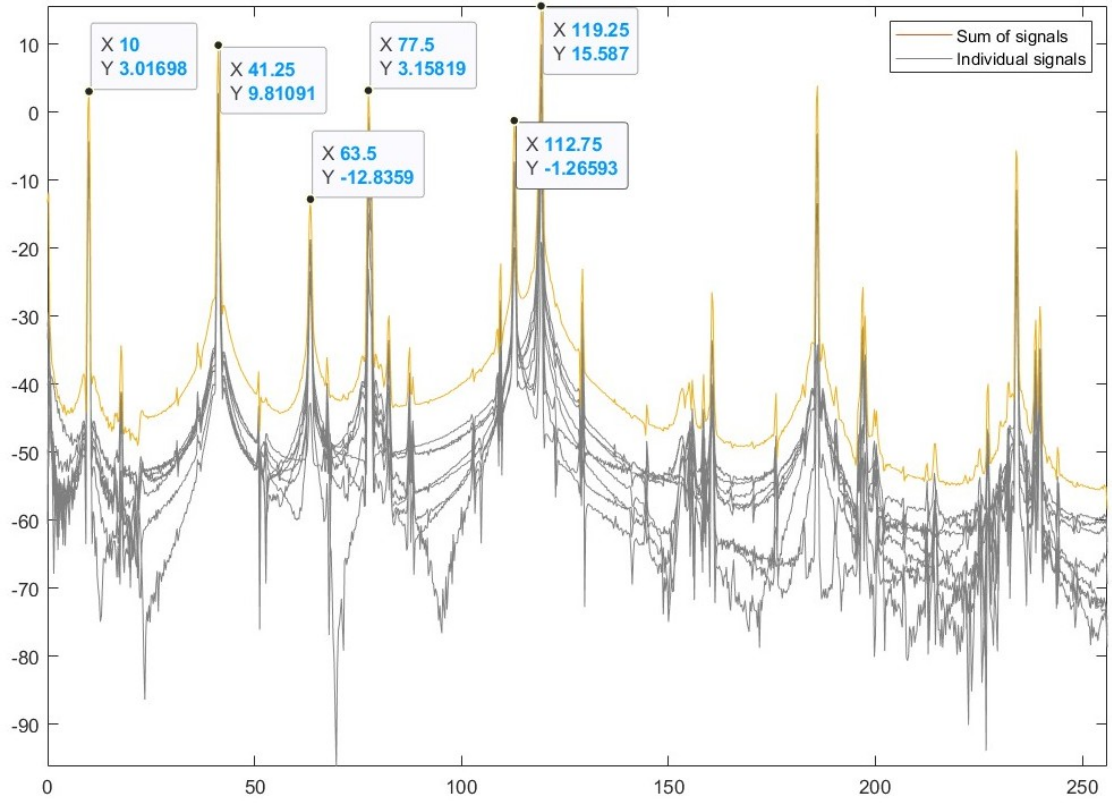


Figure 3: Summed auto-PSD across all acquisition channels.

The individual auto-power spectral densities (auto-PSDs) of each accelerometer signal are plotted in grey, while the orange curve represents their sum. Peaks in the summed spectrum clearly indicate dominant resonant frequencies of the structure, where multiple sensors simultaneously register high vibrational energy. This summation enhances the visibility of global modes and helps identify natural frequencies more robustly.

3. Stabilization Diagram

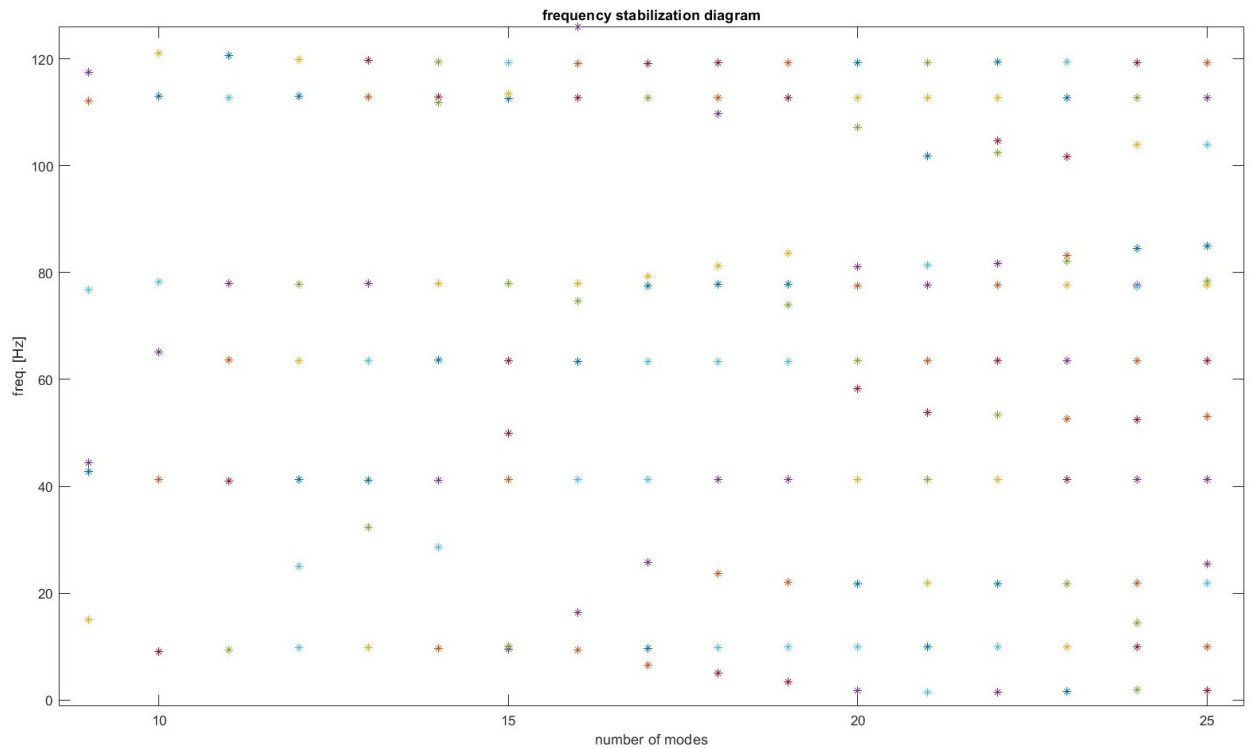
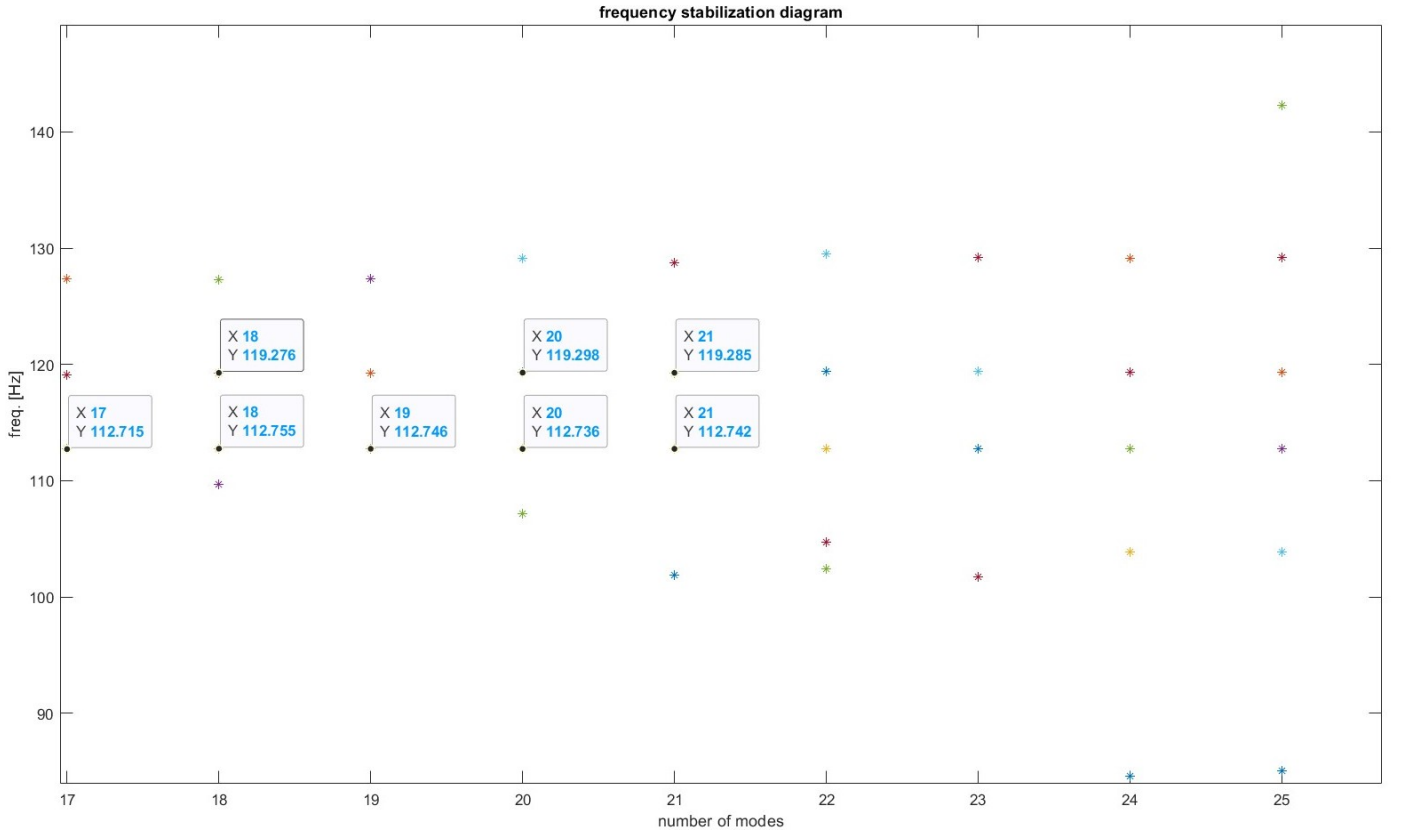
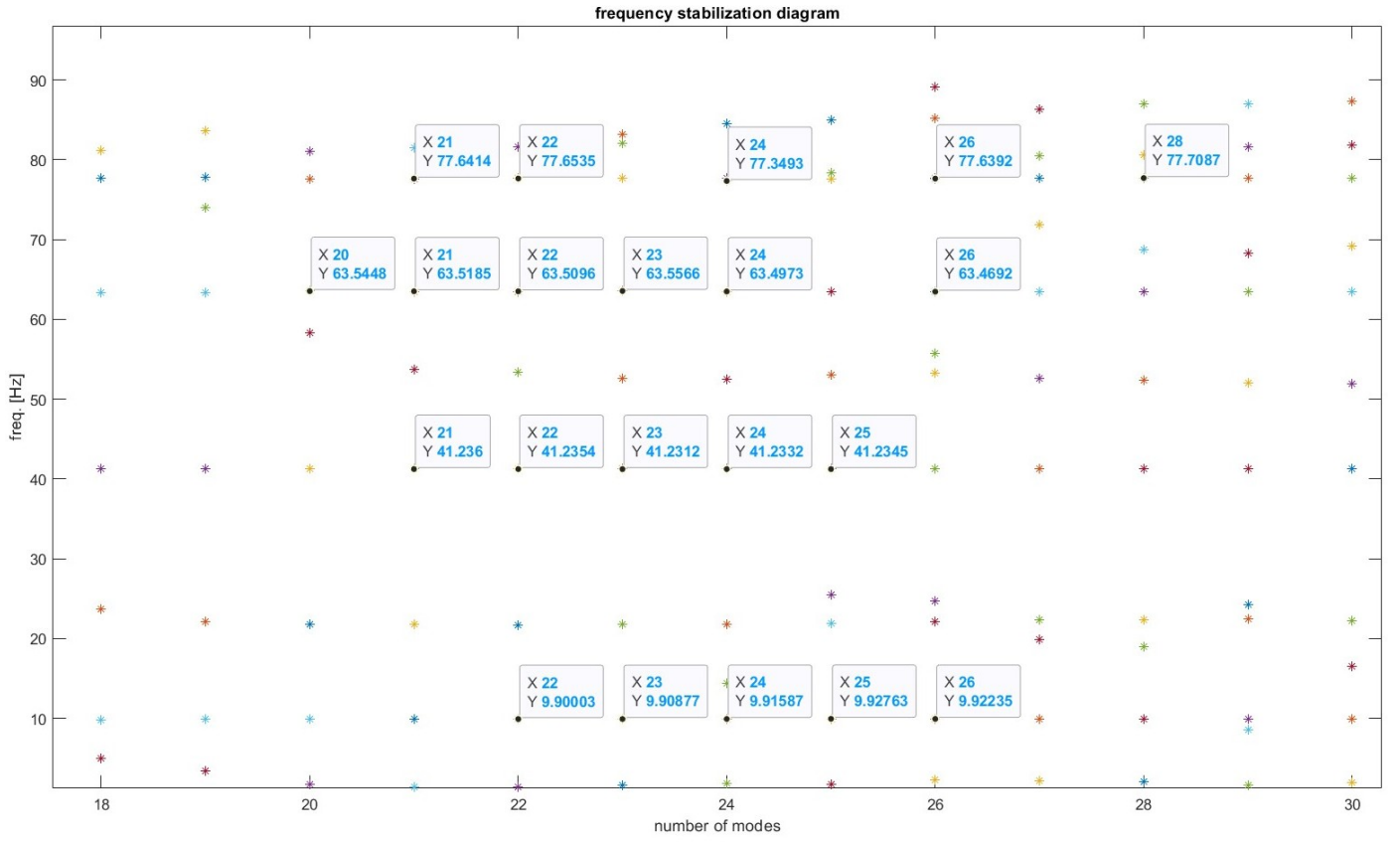


Figure 4: Stabilization diagram of channel 5.



From the stabilization diagram, we can identify the modes once again, and we are able to pick the frequencies and check their stability along the number of modes.

4. Modal Parameters

Table 1: Natural frequencies and damping ratios (first 3 flexural modes).

Mode	Frequency (Hz)	Damping (%)
1st	9.95	0.829
2nd	41.25	0.234
3rd	77.50	0.105

Comment: Estimated from stabilization + complex exponential method. There were also other modes like 63Hz, 112 Hz, but they were torsional.

5. Flexural Mode Shapes

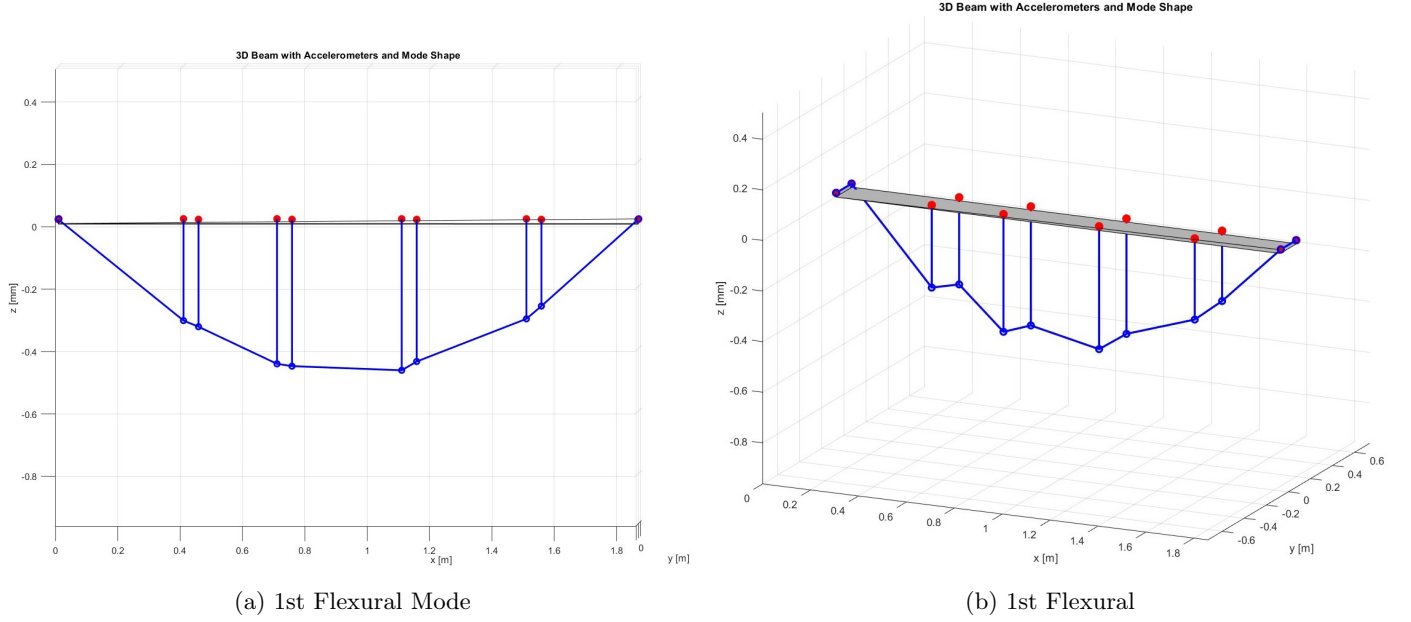


Figure 5: Identified flexural mode shapes from modal analysis.

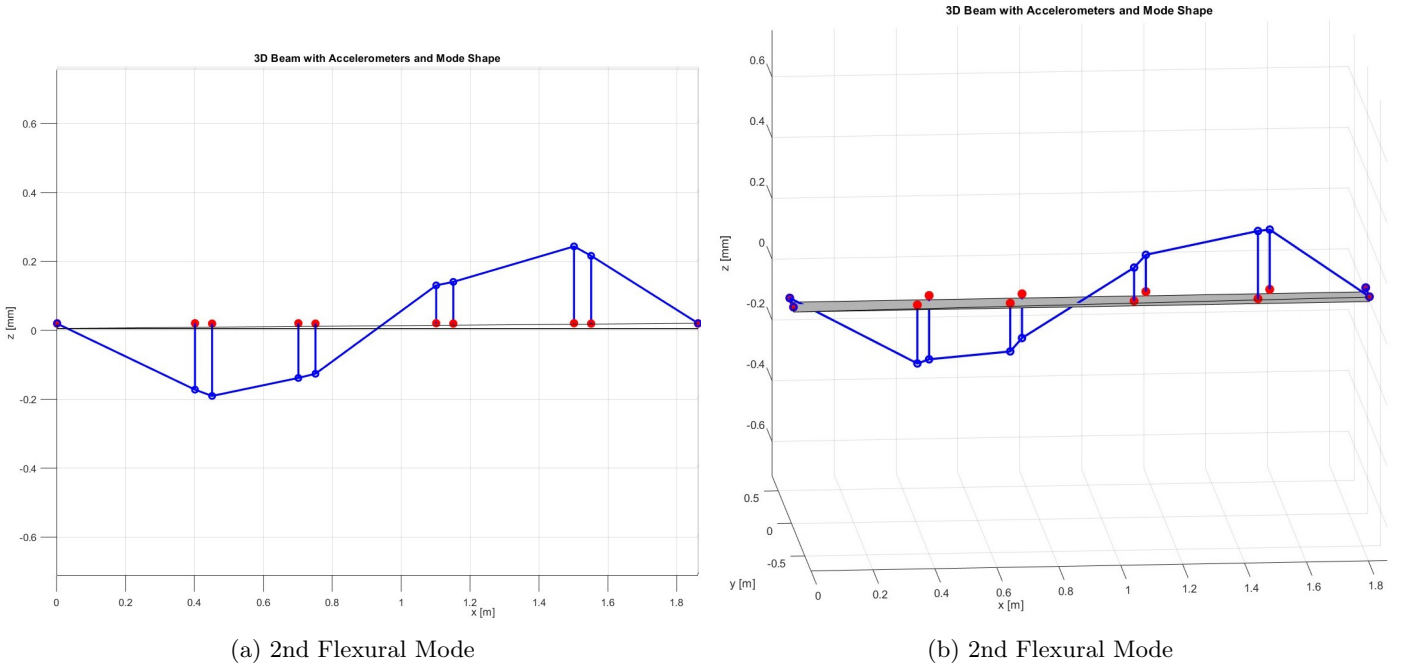


Figure 6: Identified flexural mode shapes from modal analysis.

Comment: Spatial shapes match expected structural behavior.